Samuel Francis Wallace

Event-based Facial Tracking

Master Thesis

Robotics and Perception Group
University of Zurich

Supervision

Roberto Pellerito
Nikola Zubić
Prof. Dr. Davide Scaramuzza

March 2026

Contents

Abstract **iii**

Nomenclature **v**

1 Introduction **1**
1.1 Motivation . . . 1
1.2 Problem Statement . . . 2
1.3 Approach Overview . . . 2
1.4 Contributions . . . 3
1.5 Related Work . . . 3
1.5.1 3D Morphable Face Models . . . 3
1.5.2 Event-based Point Tracking . . . 3
1.5.3 Frame Interpolation . . . 3
1.5.4 Facial Performance Capture . . . 4
1.5.5 Event-based 3D Reconstruction . . . 4
1.6 Thesis Outline . . . 4

2 Background **6**
2.1 Event Cameras . . . 6
2.2 FLAME: 3D Morphable Face Model . . . 6
2.2.1 Historical Context . . . 7
2.3 ETAP: Event-based Tracking of Any Point . . . 7
2.3.1 Event Representation . . . 7
2.3.2 Architecture . . . 8
2.3.3 Training Losses . . . 8
2.4 MediaPipe Face Landmarker . . . 8
2.5 Monocular Face Reconstruction . . . 8
2.5.1 DECA . . . 8
2.5.2 MICA . . . 9
2.6 Frame Interpolation Baseline . . . 10
2.6.1 TimeLens . . . 10
2.6.2 TimeLens++ and TimeLens-XL . . . 10

3 Methodology **12**
3.1 Overview . . . 14
3.2 Preprocessing . . . 14
3.2.1 MediaPipe Landmark Detection . . . 14

3.2.2 ETAP Landmark Tracking . . . 14
3.2.3 B-spline Fitting . . . 15
3.3 Canonical Shape Estimation . . . 16
3.4 Pass 1: Keyframe Fitting . . . 16
3.4.1 Stage 1: Rigid Camera Pose . . . 16
3.4.2 Stage 2: Joint Photometric and Landmark Fitting . . . 16
3.5 Pass 2: Intermediate Frame Fitting . . . 17
3.5.1 Camera Pose Interpolation . . . 17
3.5.2 Optimisation . . . 17
3.6 Contrast Maximisation . . . 18
3.6.1 Image of Warped Events . . . 18
3.6.2 Flow Computation . . . 18
3.6.3 Event Coordinate Transform . . . 20
3.6.4 Loss Formulation . . . 21
3.7 Implementation Details . . . 21

4 Experiments **22**
4.1 Evaluation Protocol . . . 22
4.1.1 Data . . . 22
4.1.2 Held-out Frame Protocol . . . 22
4.1.3 Metrics . . . 23
4.1.4 Methods . . . 24
4.2 FAN Landmark Accuracy . . . 24
4.3 TimeLens Analysis . . . 25
4.3.1 Frame Coverage . . . 25
4.3.2 Temporal Coherence . . . 26
4.4 Rapid Blink: Eye Dynamics . . . 26
4.4.1 Per-Event Analysis . . . 27
4.4.2 Cross-Subject EAR . . . 27
4.5 Rapid Head Motion . . . 27
4.5.1 Per-Interval Analysis . . . 28
4.5.2 Qualitative Results . . . 28
4.6 Temporal Properties . . . 28
4.6.1 Expression Velocity . . . 28
4.6.2 Expression Dynamic Range . . . 29
4.7 Contrast Maximisation . . . 29
4.7.1 IWE Variance . . . 30
4.7.2 CM Without Regularisation . . . 30

5 Discussion **33**
5.1 Conclusion . . . 33
5.2 Future Work . . . 33

A Something **35**

Abstract

We present Event-based Facial Tracking (EFT), a method that fits the FLAME 3D morphable face model at high temporal resolution by combining sparse RGB keyframes with event camera data. EFT tracks facial landmarks between keyframes using ETAP (Event-based Tracking of Any Point), smooths the trajectories with B-splines, and optimises FLAME expression parameters through a two-pass fitting architecture that separates camera pose from expression estimation.

We evaluate on eight sequences spanning standard and rapid facial motions, comparing EFT against linear interpolation, TimeLens (event-based frame interpolation), and two contrast maximisation variants. EFT is the only event-based method that produces output on 100% of frames: TimeLens fails to reconstruct 50–63% of intermediate frames, and the frames it does reconstruct exhibit severe temporal jitter. On rapid motions, EFT captures dynamics that linear interpolation misses: it reduces FAN landmark error by 31% on fast head rotation and EAR error by 16% on rapid blinking. EFT also produces the smoothest expression trajectories of all methods on standard sequences, while maintaining expression dynamic range closest to the ground truth on rapid sequences.

We further show that contrast maximisation alone—without landmark supervision or expression regularisation—can fit a 100-dimensional FLAME expression vector from events, demonstrating that the physics of event generation implicitly constrains facial reconstruction. Our results establish event cameras as a viable sensing modality for high-temporal-resolution 3D facial tracking.

Nomenclature

Notation

$\mathbf{J}$	Jacobian
$\mathbf{H}$	Hessian
$\mathbf{T}_{WB}$	coordinate transformation from frame B to frame W
$\mathbf{R}_{WB}$	orientation of B with respect to W
${}_W\mathbf{t}_{WB}$	translation of B with respect to W, expressed in coordinate system W

Scalars are written in lower case letters (a), vectors in lower case bold letters ($\mathbf{a}$) and matrices in upper case bold letters ($\mathbf{A}$).

Acronyms and Abbreviations

RPG	Robotics and Perception Group
DoF	Degree of Freedom
IMU	Inertial Measurement Unit
MAV	Micro Aerial Vehicle
ROS	Robot Operating System

Chapter 1

Introduction

High-temporal-resolution 3D facial tracking enables applications in animation, virtual reality, and expression analysis. We present Event-based Facial Tracking (EFT), a method that leverages event cameras to achieve accurate 3D facial tracking at high temporal resolution without requiring dense RGB frames. Our approach combines sparse RGB keyframes with event-based point tracking to fit the FLAME 3D morphable face model at intermediate timesteps. We demonstrate that EFT maintains consistent accuracy across increasing temporal sparsity, while neural frame interpolation methods degrade significantly.

1.1 Motivation

High-fidelity facial performance capture is central to modern film and game production. State-of-the-art s ystems such as Medusa [1], developed by Disney Research and operated by Industrial Light & Magic, reconstruct facial geometry at pore-level detail and have been used in major productions including the Star Wars and Marvel franchises. However, Medusa and similar systems require dense multi-camera rigs in controlled studio environments, making them expensive and inaccessible outside large studios. More recent work from Disney Research [23] explores neural representations for relightable and animatable digital heads, but these methods still rely on specialized capture hardware.

Standard frame-based facial tracking methods offer a more accessible alternative but are fundamentally limited by camera frame rate. Applications such as facial performance capture for animation require high temporal resolution to capture rapid expressions and subtle muscle movements. Increasing frame rate introduces practical constraints: higher bandwidth, storage requirements, and computational cost. Furthermore, motion blur at high speeds degrades tracking quality even when high frame rates are available.

Event cameras offer an alternative sensing paradigm. These neuromorphic sensors asynchronously report per-pixel brightness changes with microsecond temporal resolution. Event cameras consume low power, produce sparse output, and do not suffer from motion blur. These properties make event cameras well-suited

for tracking fast facial movements, potentially reducing the need for expensive multi-camera rigs by providing high temporal resolution from a single sensor.

The challenge lies in combining the geometric accuracy of frame-based 3D face model fitting with the temporal resolution of event cameras. Frame-based methods fit 3D morphable models like FLAME to detected landmarks with high accuracy. Event-based methods can track points at high temporal resolution but lack the geometric constraints for 3D reconstruction. We bridge this gap by using sparse RGB keyframes for robust FLAME fitting and event-based tracking for temporal interpolation.

1.2 Problem Statement

Given a synchronized stream of RGB frames and events from a hybrid camera system, we aim to reconstruct 3D facial geometry at temporal resolutions exceeding the RGB frame rate. We formulate this as fitting FLAME model parameters at intermediate timesteps between sparse RGB keyframes.

Let $\mathbf{I}_k$ denote the RGB frame at keyframe index k, and let $\mathbf{E}_{k\to k+1}$ denote the events occurring between consecutive keyframes. Our goal is to estimate FLAME parameters $\boldsymbol{\theta}_t \in \mathbb{R}^{106}$ for intermediate timesteps $t \in (k, k+1)$ such that the reconstructed face mesh accurately represents the facial geometry at time t.

The key technical challenges are:

1. Establishing dense temporal correspondences between keyframes using events.
2. Fitting 3D face model parameters to sparse landmark observations.
3. Maintaining geometric consistency across the interpolated sequence.

1.3 Approach Overview

EFT addresses these challenges through a three-stage pipeline. First, we detect facial landmarks on RGB keyframes using MediaPipe [14] and initialize FLAME [12] parameters using DECA [6] for expression and pose, and MICA [24] for identity shape. Second, we track these landmarks through event data using ETAP[9] (Event-based Tracking of Any Point), a neural network that propagates point positions using event streams. Third, we fit B-splines to the tracked landmark trajectories and optimize FLAME expression parameters at intermediate timesteps.

Our two-pass fitting architecture separates camera pose estimation from expression optimization. In the first pass, we fit keyframes by jointly optimizing camera pose and expression parameters using detected landmarks. In the second pass, we fit intermediate frames by optimizing only expression parameters while interpolating camera pose from flanking keyframes. This separation improves stability and reduces drift in the reconstructed sequence.

1.4 Contributions

This thesis makes the following contributions:

1. We present EFT, a method for high-temporal-resolution 3D facial tracking that combines event cameras with the FLAME 3D morphable face model.
2. We introduce a two-pass fitting architecture that separates keyframe fitting from intermediate frame fitting, using B-spline interpolated landmarks for temporal densification.
3. We provide a fair comparison benchmark against TimeLens, a neural frame interpolation baseline, using 2D landmark reprojection error as the evaluation metric. VERIFY
4. We demonstrate that EFT maintains consistent accuracy (NME ≈ 0.06) across keyframe strides from 2 to 16, while TimeLens degrades from NME 0.04 to 0.15 and exhibits up to 40% failure rate at stride 8.

1.5 Related Work

1.5.1 3D Morphable Face Models

The concept of 3D morphable face models (3DMMs) was introduced by Blanz and Vetter [2], who represented faces as linear combinations of basis shapes learned from 3D scans. The SMPL body model [13] later introduced a parametric formulation for human bodies, and FLAME [12] adapted this to the head, factorizing facial variation into identity, pose, and expression parameters. DECA [6] and MICA [24] fit FLAME to monocular images, providing real-time shape and expression estimation. However, these methods operate per-frame and do not model temporal consistency across video sequences.

1.5.2 Event-based Point Tracking

ETAP [9] extends point tracking to event cameras, achieving state-of-the-art performance on event-based tracking benchmarks. ETAP outperforms frame-based methods when motion blur or low frame rate degrades RGB tracking quality.

1.5.3 Frame Interpolation

TimeLens [20] introduced event-based video frame interpolation (VFI), combining warping-based and synthesis-based interpolation through an attention-based fusion module. TimeLens++ [21] replaced the image-level fusion with multi-scale feature-level fusion and introduced a motion spline estimator for temporally consistent flow. TimeLens-XL [15] further improved efficiency and large-motion handling by decomposing motion into multi-step small displacements with an iterative trajectory estimator, achieving real-time performance with only 4M parameters. However, all TimeLens variants operate in pixel space and produce artifacts when the temporal gap between input frames is large. Our experiments show that TimeLens degrades significantly at high keyframe strides, while geometric tracking methods like EFT maintain accuracy.

1.5.4 Facial Performance Capture

The Medusa system [1], developed at Disney Research, reconstructs facial geometry at pore-level resolution using a mobile multi-camera rig without motion-capture markers. Medusa has been used extensively in film production and was recognized with an Academy Scientific and Technical Award in 2019. Xu et al. [23] build on this line of work with neural relightable avatars, but these methods still require controlled studio environments and specialized hardware. Monocular methods trade accuracy for accessibility, yet most event-based facial methods focus on 2D landmark detection rather than 3D reconstruction. EFT bridges this gap by using event-based tracking to guide 3D morphable model fitting, offering high temporal resolution from a lightweight sensor setup.

1.5.5 Event-based 3D Reconstruction

Recent work has applied event cameras to 3D human body reconstruction, but these methods are fundamentally unsuited for facial tracking. EventCap [22] performs monocular 3D capture of high-speed human motions at up to 1000 fps by combining an event camera with a single RGB camera. EventCap uses the SMPL body model [13] and optimizes a skeletal pose $\mathbf{S} = [\boldsymbol{\theta}, \mathbf{R}, \mathbf{t}]$ with 27 joint angles through a three-stage pipeline: event trajectory generation, hybrid batch pose optimization, and event-based pose refinement via ICP on silhouette contours. However, EventCap is designed for large-scale articulated body motion and relies on silhouette-based refinement that assumes distinct body contours. Facial expressions involve sub-centimetre non-rigid surface deformations that do not produce silhouette changes, making EventCap's approach inapplicable. Furthermore, EventCap uses SMPL, which does not model facial geometry—the head region contains too few vertices to capture expression.

Kohyama et al. [11] propose event-only 3D human body reconstruction using a moving event camera that orbits a stationary subject. Their method classifies event contours with a CNN, performs voxel carving at 256^3 resolution, and fits SMPL via Chamfer distance minimization. However, the authors explicitly note that their voxel resolution "cannot reproduce fine structures like faces." Moreover, the method assumes a static subject and a moving camera—the opposite of our setting where the subject moves and the camera is fixed. As with EventCap, the reliance on SMPL and external silhouettes precludes application to facial expression tracking.

These methods demonstrate the potential of event cameras for high-speed 3D reconstruction but highlight a gap: no existing event-based method addresses 3D facial tracking with a morphable face model. EFT fills this gap by combining event-based point tracking with the FLAME face model.

1.6 Thesis Outline

The remainder of this thesis is organized as follows.

Chapter 2 provides background on the FLAME model, event cameras, ETAP, and the monocular reconstruction methods used in our pipeline.

Chapter 3 describes the EFT method in detail, including the two-pass fitting architecture and B-spline landmark interpolation.

Chapter 4 evaluates EFT against baselines using 2D landmark reprojection error and 3D mesh metrics.

Chapter 5 discusses limitations, failure cases, and directions for future work.

Chapter 2

Background

This chapter provides the technical foundations for the components used in our pipeline. We first describe the two core elements of EFT: event cameras (Section 2.1), which provide high-temporal-resolution motion data, and the FLAME 3D morphable face model (Section 2.2), which parameterises facial geometry. We then describe the methods that connect them: ETAP for event-based point tracking (Section 2.3), MediaPipe for landmark detection (Section 2.4), and DECA and MICA for monocular face reconstruction (Section 2.5). Finally, we describe TimeLens (Section 2.6), the frame interpolation method used as our primary baseline.

2.1 Event Cameras

Event cameras are neuromorphic sensors that asynchronously report per-pixel brightness changes with microsecond temporal resolution. Unlike frame-based cameras, which capture entire frames at fixed intervals, event cameras independently monitor each pixel and generate an event whenever the log-intensity change at that pixel exceeds a contrast threshold C. Formally, an event $e_i = (x_i, y_i, t_i, p_i)$ is triggered at pixel (x_i, y_i) and time t_i when:

$$|\log I(x_i, y_i, t_i) - \log I(x_i, y_i, t_i - \Delta t)| \geq C, \tag{2.1}$$

where $I(x, y, t)$ is the image intensity and the polarity $p_i \in \{-1, +1\}$ indicates the sign of the brightness change.

This asynchronous, sparse output results in high dynamic range ($> 120\,\mathrm{dB}$), no motion blur, and low power consumption. These properties make event cameras well-suited for tracking fast motion in low-latency applications.

2.2 FLAME: 3D Morphable Face Model

FLAME [12] (Faces Learned with an Articulated Model and Expressions) is a parametric 3D head model that factorizes facial variation into identity shape parameters $\boldsymbol{\beta} \in \mathbb{R}^{|\boldsymbol{\beta}|}$, pose parameters $\boldsymbol{\theta} \in \mathbb{R}^{3K+3}$ (axis-angle rotations for $K =$

4 joints plus global rotation), and expression parameters $\boldsymbol{\psi} \in \mathbb{R}^{|\boldsymbol{\psi}|}$. FLAME extends the SMPL body model formulation [13] to the head by adding four joints (neck, jaw, and two eyeballs) and an expression blendshape space.

The FLAME mesh $M(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\psi}) \in \mathbb{R}^{n \times 3}$ with $n = 5023$ vertices is computed as:

$$M(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\psi}) = W(T_P(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\psi}),\ \mathbf{J}(\boldsymbol{\beta}),\ \boldsymbol{\theta},\ \mathcal{W}), \tag{2.2}$$

where $W(\cdot)$ denotes the linear blend skinning (LBS) function, $\mathbf{J}(\boldsymbol{\beta}) \in \mathbb{R}^{K \times 3}$ are the joint locations regressed from the identity shape, and $\mathcal{W} \in \mathbb{R}^{n \times K}$ are the skinning weights. In LBS, each mesh vertex is assigned skinning weights that specify how much each joint influences its position; the final vertex location is computed as a weighted sum of the vertex transformed by each joint's rotation matrix.

The posed template T_P is obtained by adding identity, pose-dependent, and expression corrective blendshapes to the mean template $\bar{\mathbf{T}} \in \mathbb{R}^{n \times 3}$:

$$T_P(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\psi}) = \bar{\mathbf{T}} + B_S(\boldsymbol{\beta}) + B_P(\boldsymbol{\theta}) + B_E(\boldsymbol{\psi}), \tag{2.3}$$

where $B_S(\boldsymbol{\beta}) = \sum_{i=1}^{|\boldsymbol{\beta}|} \beta_i \mathbf{S}_i$ applies identity shape blendshapes, $B_P(\boldsymbol{\theta})$ applies pose-dependent corrective blendshapes, and $B_E(\boldsymbol{\psi}) = \sum_{i=1}^{|\boldsymbol{\psi}|} \psi_i \mathbf{E}_i$ applies expression blendshapes. The identity basis $\{\mathbf{S}_i\}$ was trained on approximately 3800 head scans from the CAESAR dataset, while the expression basis $\{\mathbf{E}_i\}$ was learned from the D3DFACS dataset of 4D face sequences.

2.2.1 Historical Context

FLAME builds on a line of work in 3D morphable face models (3DMMs). The concept of 3D morphable face models was introduced by Blanz and Vetter [2], who represented faces as linear combinations of basis shapes learned via PCA on aligned 3D head scans. The Basel Face Model (BFM) [17] extended this to a larger scan corpus, improving generalization. However, these early models lack articulated joints and do not disentangle expression from identity. The SMPL body model [13] later introduced a learned parametric formulation combining an identity shape space, an articulated skeleton, and pose-dependent corrective blendshapes. SMPL became the dominant representation for human body modelling but does not model the face, as the head region contains too few vertices to capture facial motion. FLAME adapts the SMPL formulation to the head, addressing this limitation.

2.3 ETAP: Event-based Tracking of Any Point

ETAP [9] (Event-based Tracking of Any Point) extends the TAP-Vid [5] point tracking framework to event cameras.

2.3.1 Event Representation

ETAP converts the asynchronous event stream into mixed-density event stacks for processing by a convolutional network. Given events between two timesteps, the stream is divided into $B = 10$ temporal bins using a hierarchical binning

strategy that preserves fine temporal resolution near the target timestep while aggregating events further away. Each bin accumulates the sum of event polarities per pixel, producing a tensor $\mathbf{V} \in \mathbb{R}^{B \times H \times W}$.

2.3.2 Architecture

ETAP processes events in a sliding window of $w = 8$ timesteps with stride $T_s = 4$. For each query point, the network extracts multi-scale correlation features at $S = 4$ spatial scales by computing the inner product between the query point feature and features at candidate locations around the predicted position.

Position estimates are refined iteratively through M rounds of a transformer block γ that applies alternating intra-point attention (across the temporal dimension for each point) and temporal attention (across points at each timestep):

$$(\hat{\mathbf{p}}^{(m+1)}, \hat{v}^{(m+1)}) = \gamma\Big(\hat{\mathbf{p}}^{(m)}, \hat{v}^{(m)}, \mathbf{F}_{\text{corr}}\Big), \quad m = 1, \ldots, M, \tag{2.4}$$

where $\hat{\mathbf{p}}^{(m)}$ and $\hat{v}^{(m)}$ are the position and visibility estimates at iteration m, and $\mathbf{F}_{\text{corr}}$ are the multi-scale correlation features. ETAP uses $M = 4$ iterations during training and $M = 6$ during evaluation.

2.3.3 Training Losses

ETAP is trained with a combination of three losses:

$$\mathcal{L} = 0.1 \cdot \mathcal{L}_{\text{tp}} + \mathcal{L}_{\text{vis}} + 0.1 \cdot \mathcal{L}_{\text{fa}}, \tag{2.5}$$

where $\mathcal{L}_{\text{tp}}$ is the trajectory prediction loss (Huber loss on predicted positions), $\mathcal{L}_{\text{vis}}$ is a binary cross-entropy visibility loss, and $\mathcal{L}_{\text{fa}}$ is a feature alignment loss that encourages motion-invariant features via temporal inversion—the features extracted when tracking forward in time should match those when tracking backward.

2.4 MediaPipe Face Landmarker

MediaPipe [14] provides a real-time face landmark detection model that outputs 478 dense facial landmarks from a single RGB image. We use these detections as the ground-truth landmark observations on RGB keyframes.

2.5 Monocular Face Reconstruction

DECA and MICA provide initialisation for FLAME parameters in our pipeline. DECA estimates expression and pose from a single RGB image, while MICA estimates identity shape.

2.5.1 DECA

DECA [6] (Detailed Expression Capture and Animation) regresses FLAME parameters from a single RGB image using a coarse-to-detail architecture.

The coarse stage uses a ResNet-50 encoder E_c that maps an input image $\mathbf{I}$ to a 236-dimensional latent code:

$$E_c(\mathbf{I}) = (\boldsymbol{\beta}, \boldsymbol{\psi}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \mathbf{l}, \mathbf{c}), \tag{2.6}$$

where $\boldsymbol{\beta} \in \mathbb{R}^{100}$ are identity shape parameters, $\boldsymbol{\psi} \in \mathbb{R}^{50}$ are expression parameters, $\boldsymbol{\theta} \in \mathbb{R}^{6}$ encodes jaw pose and global rotation, $\boldsymbol{\alpha} \in \mathbb{R}^{50}$ are albedo coefficients, $\mathbf{l} \in \mathbb{R}^{27}$ are spherical harmonics illumination coefficients (9 bands $\times$ 3 colour channels), and $\mathbf{c} \in \mathbb{R}^{3}$ are camera parameters.

The coarse stage is trained with a combination of losses:

$$\mathcal{L}_{\text{coarse}} = \mathcal{L}_{\text{lmk}} + \mathcal{L}_{\text{eye}} + \mathcal{L}_{\text{pho}} + \mathcal{L}_{\text{id}} + \mathcal{L}_{\text{sc}} + \mathcal{L}_{\text{reg}}, \tag{2.7}$$

where $\mathcal{L}_{\text{lmk}}$ penalizes 2D landmark reprojection error, $\mathcal{L}_{\text{eye}}$ adds a closure-aware term for eye landmarks, $\mathcal{L}_{\text{pho}}$ is a photometric loss, $\mathcal{L}_{\text{id}}$ enforces identity consistency between different images of the same person, $\mathcal{L}_{\text{sc}}$ is a shape consistency loss, and $\mathcal{L}_{\text{reg}}$ regularizes the parameters.

The detail stage adds a secondary encoder F_d that predicts a UV-space displacement map $\mathbf{D} \in [-0.01, 0.01]^{d \times d}$ conditioned on the detail code $\boldsymbol{\delta}$, expression $\boldsymbol{\psi}$, and jaw pose $\boldsymbol{\theta}_{\text{jaw}}$:

$$\mathbf{D} = F_d(\boldsymbol{\delta}, \boldsymbol{\psi}, \boldsymbol{\theta}_{\text{jaw}}). \tag{2.8}$$

The detail displacement is applied to the coarse mesh normals in UV space: $\mathbf{M}'_{\text{uv}} = \mathbf{M}_{\text{uv}} + \mathbf{D} \odot \mathbf{N}_{\text{uv}}$, producing a detailed normal map for rendering. In our pipeline, we use only the coarse stage of DECA for expression and pose initialization.

2.5.2 MICA

MICA [24] (Metrical Reconstruction of Human Faces) improves identity shape estimation by leveraging a pre-trained face recognition network.

MICA passes the input image through an ArcFace [4] encoder (ResNet-100 trained for face recognition) to extract an identity-discriminative feature vector. A mapping network $\mathcal{M}$ consisting of three fully connected layers with ReLU activations transforms this feature into a shape code $\mathbf{z} \in \mathbb{R}^{300}$. The geometry decoder reconstructs the 3D face shape as:

$$G_{\text{3DMM}}(\mathbf{z}) = \mathbf{B} \cdot \mathbf{z} + \bar{\mathbf{A}}, \tag{2.9}$$

where $\mathbf{B}$ is the FLAME identity shape basis and $\bar{\mathbf{A}}$ is the mean shape.

The model is trained with a region-weighted vertex loss:

$$\mathcal{L} = \sum_i \left| \kappa_i \left(G_{\text{3DMM}}(\mathcal{M}(\text{ArcFace}(\mathbf{I}))) - \mathbf{G}_i^* \right) \right|, \tag{2.10}$$

where $\mathbf{G}^*$ is the ground-truth geometry and κ_i are per-region weights that emphasize the face interior ($\kappa = 150$) over the back of the head ($\kappa = 1$) and de-emphasize noisy regions such as eyes and ears ($\kappa = 0.01$). MICA was trained on 2315 subjects with pseudo-ground-truth meshes. Only the last three ResNet blocks of ArcFace are fine-tuned, preserving learned identity features. In our pipeline, MICA provides the identity shape parameters $\boldsymbol{\beta}$ for FLAME fitting.

2.6 Frame Interpolation Baseline

We use the TimeLens family of event-based video frame interpolation (VFI) methods as our primary baseline.

2.6.1 TimeLens

TimeLens [20] introduced event-based VFI through a four-module architecture. Given boundary keyframes $\mathbf{I}_0$, $\mathbf{I}_1$ and inter-frame events, the method estimates an intermediate frame $\mathbf{I}_\tau$ at arbitrary time $\tau \in (0, 1)$.

Events are converted to voxel grid representations $\mathbf{V}_{0\to\tau}$ and $\mathbf{V}_{\tau\to 1}$. A *warping module* estimates optical flow $\mathbf{F}_{\tau\to 0}$ and $\mathbf{F}_{\tau\to 1}$ from event voxel grids and warps the keyframes to time τ:

$$\mathbf{I}^{\text{warp}}_{0\to\tau} = \text{Warp}(\mathbf{I}_0, \mathbf{F}_{\tau\to 0}), \quad \mathbf{I}^{\text{warp}}_{1\to\tau} = \text{Warp}(\mathbf{I}_1, \mathbf{F}_{\tau\to 1}). \tag{2.11}$$

A *synthesis module* directly regresses a frame $\mathbf{I}^{\text{syn}}_\tau$ from the keyframes and event voxel grids, handling illumination changes and non-rigid motion that violate the brightness constancy assumption. A *refinement module* estimates residual flow $\Delta\mathbf{F}$ to correct warping errors and inpaint occluded regions. Finally, an *attention module* computes pixel-wise softmax weights $(\mathbf{W}_0, \mathbf{W}_1, \mathbf{W}_{\text{syn}})$ to blend the refined warped frames and the synthesised frame:

$$\mathbf{I}_\tau = \mathbf{W}_0 \odot \mathbf{I}^{\text{refine}}_{0\to\tau} + \mathbf{W}_1 \odot \mathbf{I}^{\text{refine}}_{1\to\tau} + \mathbf{W}_{\text{syn}} \odot \mathbf{I}^{\text{syn}}_\tau. \tag{2.12}$$

Each module uses an hourglass encoder–decoder with skip connections and is trained sequentially with a combination of ℓ_1 and perceptual losses. TimeLens degrades up to 1.6 dB PSNR when going from 1 to 15 skipped frames, compared to up to 5.4 dB for frame-only methods.

2.6.2 TimeLens++ and TimeLens-XL

TimeLens++ [21] addresses three limitations of TimeLens: (1) image-level fusion that propagates synthesis artifacts, (2) per-frame flow estimation at $O(N)$ cost, and (3) events-only motion estimation that fails in low-contrast regions. TimeLens++ replaces image-level blending with *multi-scale feature-level fusion* using gated compression modules, introduces a *motion spline estimator* that fits cubic splines with K control points per pixel from both events and images, and resamples flow at any time τ in $O(1)$. The model uses 53.9M parameters compared to TimeLens's 72.2M.

TimeLens-XL [15] further improves efficiency by decomposing the motion between keyframes into multi-step small displacements and recursively estimating the motion trajectory:

$$\hat{\mathbf{I}}_{t_{k+1}},\ \hat{\mathbf{f}}_{t_{k+1}\to t_k} = \Phi\Big(\{E_\tau\}_{t_k}^{t_{k+1}},\ \hat{\mathbf{I}}_{t_k},\ \hat{\mathbf{f}}_{t_k\to t_{k-1}}\Big), \tag{2.13}$$

where each step estimates both an intermediate frame and flow from locally relevant events. TimeLens-XL achieves real-time performance (26 fps at 1280×720) with only 4M parameters—roughly 1/20 of TimeLens.

Despite these advances, all TimeLens variants operate in pixel space and synthesise appearance rather than recovering 3D geometry. When the temporal gap between keyframes grows large, synthesis quality degrades and landmark detection on interpolated frames becomes unreliable. We use TimeLens as our baseline to demonstrate that geometric model fitting via event-based tracking provides more robust high-temporal-resolution facial reconstruction.

Chapter 3

Methodology

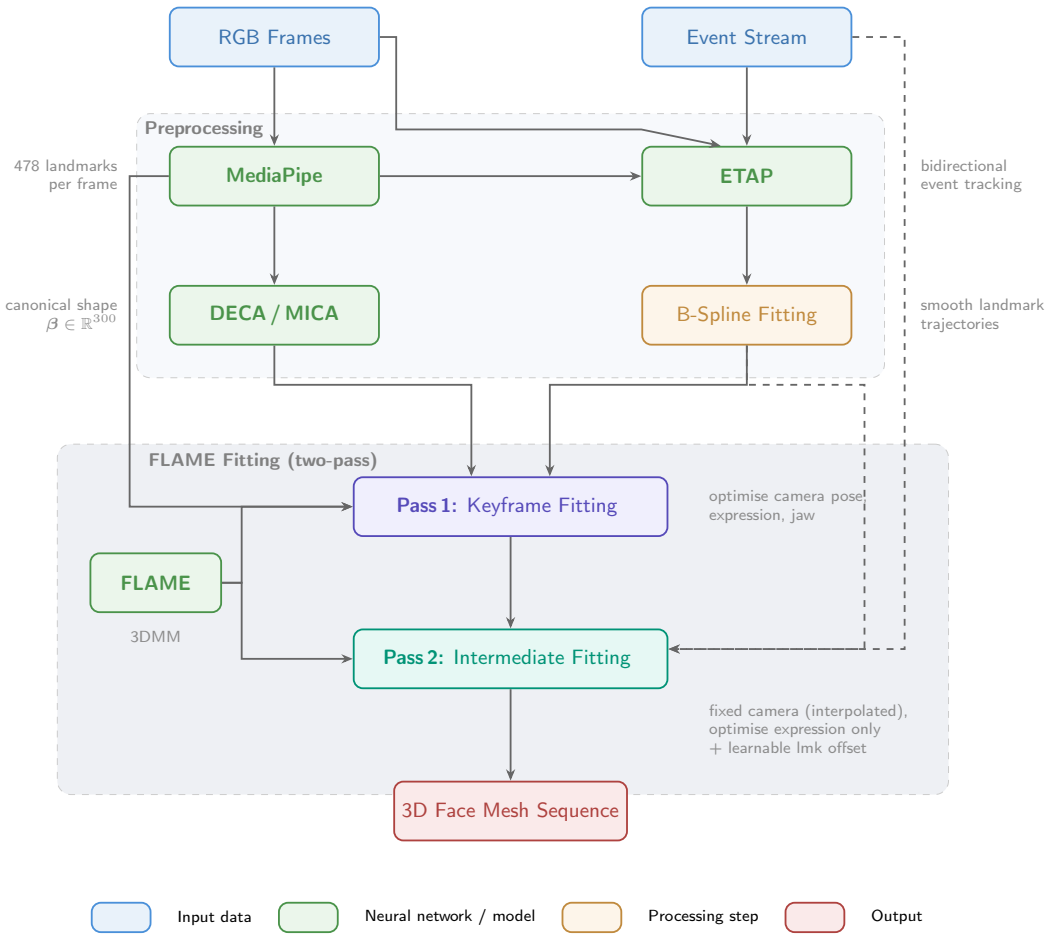


Figure 3.1: Overview of the event-based facial tracking pipeline. RGB frames are processed by MediaPipe and DECA/MICA for landmark detection and shape initialisation, while event streams are tracked by ETAP and smoothed via B-spline fitting. A two-pass FLAME fitting procedure produces the final 3D face mesh sequence.

3.1 Overview

This chapter describes the EFT pipeline in detail. We cover preprocessing (Section 3.2), canonical shape estimation (Section 3.3), the two-pass fitting architecture (Sections 3.4 and 3.5), contrast maximisation (Section 3.6), and implementation details (Section 3.7).

EFT takes as input a synchronised stream of RGB frames and events from a hybrid camera system and produces per-timestep FLAME parameters. The pipeline proceeds in three stages, as illustrated in Figure 3.1. First, we preprocess the input by detecting landmarks on RGB keyframes, tracking them through the event stream with ETAP, and fitting B-splines to the resulting trajectories. Second, we estimate a canonical identity shape from a subset of frames using MICA. Third, we fit FLAME parameters in two passes: Pass 1 fits keyframes using detected landmarks and photometric supervision, and Pass 2 fits intermediate frames using B-spline landmarks with the camera pose interpolated from flanking keyframes.

3.2 Preprocessing

Preprocessing converts the raw RGB frames and event stream into the landmark observations used by the fitting stages. We describe the three sub-stages: landmark detection on RGB keyframes, event-based landmark tracking between keyframes, and B-spline smoothing of the tracked trajectories.

3.2.1 MediaPipe Landmark Detection

We detect facial landmarks on each RGB keyframe using MediaPipe [14], which outputs 478 dense landmarks per face. We convert these to a 68-landmark subset compatible with the DLIB landmark convention, which aligns with the FLAME model's landmark definition. An additional 10 eye landmarks are extracted from MediaPipe's iris tracking indices for eye-specific fitting in the later stages.

Detection failures are logged per frame. When MediaPipe fails to detect a face on a keyframe, we fall back to DECA [6] initialisation and interpolate parameters from neighbouring successful keyframes (Section 3.4).

3.2.2 ETAP Landmark Tracking

We use ETAP [9] to track the detected MediaPipe landmarks through the event stream between consecutive keyframe pairs. The 68 facial landmarks from each keyframe serve as query points for ETAP. Events between two keyframes are converted into mixed-density voxel stacks with $B = 10$ temporal bins and 20 000 events per stack (see Section 2.3). ETAP processes these in a sliding window of 8 timesteps with 6 refinement iterations, producing position estimates at 50 intra-slice timesteps per keyframe interval.

We employ bidirectional tracking to improve robustness. The forward pass tracks landmarks from keyframe k through the event stream towards keyframe $k+1$, while the backward pass tracks from $k+1$ towards k. The two trajectories

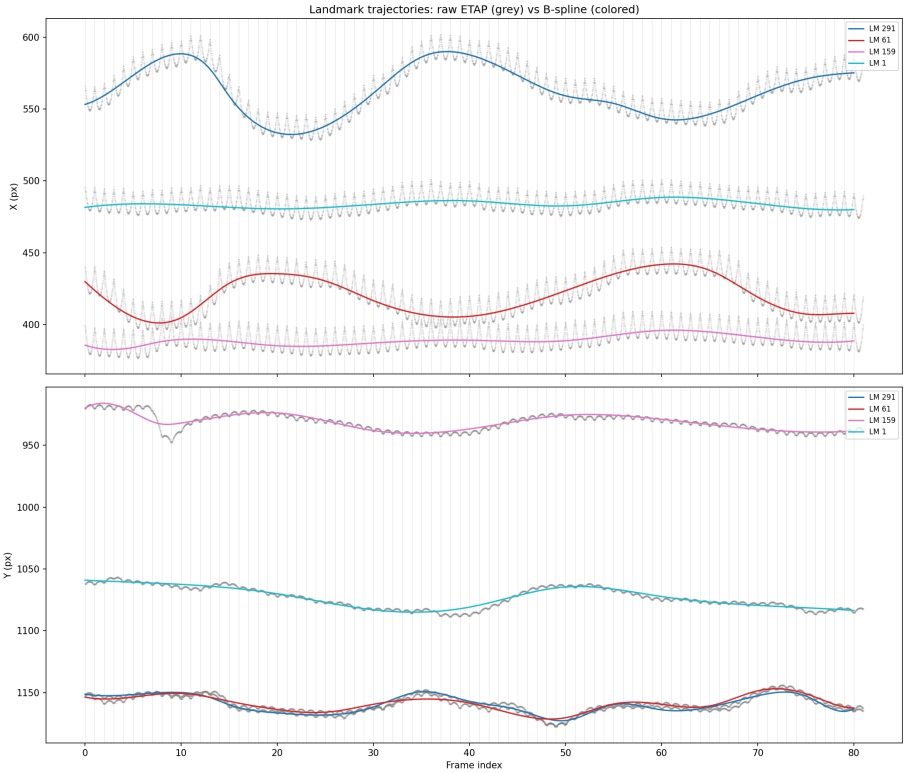


Figure 3.2: Raw ETAP landmark trajectories (grey) and fitted B-splines (coloured) for four landmarks over 80 frames. Top: x-coordinate. Bottom: y-coordinate. The B-spline preserves the underlying motion while removing high-frequency tracking noise.

are merged via linear temporal blending: at relative time $\alpha \in [0, 1]$ within the interval, the merged position is

$$\mathbf{p}(\alpha) = (1 - \alpha) \cdot \mathbf{p}_{\text{fwd}}(\alpha) + \alpha \cdot \mathbf{p}_{\text{bwd}}(\alpha). \tag{3.1}$$

This weighting trusts the forward pass more near keyframe k and the backward pass more near keyframe $k + 1$, reducing drift at the interval boundaries.

3.2.3 B-spline Fitting

The raw ETAP trajectories contain high-frequency noise from the event-based tracking process. Figure 3.2 illustrates this on four representative landmarks: the grey traces show the raw ETAP output, which exhibits a systematic saw-tooth pattern caused by ETAP's sliding-window processing, while the coloured curves show the fitted B-splines. We smooth each landmark trajectory by fitting a cubic B-spline (degree 3) with smoothing factor $s = 500$ using `scipy.interpolate.splprep`. The spline is fit to the per-frame median positions of the ETAP sub-frame estimates.

The B-spline parameterisation enables evaluation at arbitrary intermediate timesteps,

providing smooth landmark observations for Pass 2 fitting. The fitted splines are cached for efficient repeated evaluation during optimisation.

3.3 Canonical Shape Estimation

Before fitting individual frames, we estimate a canonical identity shape $\boldsymbol{\beta}$ that remains fixed across the entire sequence. MICA [24] provides the identity shape code by processing face crops through its ArcFace [4] encoder and mapping network (see Section 2.5.2).

EFT supports two modes for canonical shape estimation. In photometric mode, we sample 8 evenly-spaced frames from the sequence, fit FLAME to each, and average the resulting shape and texture parameters. In landmark-only mode, we average the MICA shape codes from the first 3 frames. Both modes ensure a consistent identity across the full sequence by preventing per-frame shape drift.

3.4 Pass 1: Keyframe Fitting

Pass 1 fits FLAME parameters to each RGB keyframe using detected landmarks and, optionally, photometric supervision. Each keyframe undergoes a two-stage optimisation that separates rigid camera pose estimation from expression and appearance fitting.

3.4.1 Stage 1: Rigid Camera Pose

The first stage estimates the 6-DoF camera pose (Euler rotation and translation) and field of view for each keyframe. All other FLAME parameters are held fixed at their initialised values.

We optimise for 1500 iterations using Adam [10] with an initial learning rate of 0.05, reduced to 0.005 at iteration 1000. The loss is a weighted landmark reprojection error:

$$\mathcal{L}_{\text{pose}} = w_{\text{face}} \cdot \mathcal{L}_{\text{face}} + w_{\text{jaw}} \cdot \mathcal{L}_{\text{jaw}} + w_{\text{ear}} \cdot \mathcal{L}_{\text{ear}} + w_{\text{temp}} \cdot \mathcal{L}_{\text{temp}}, \tag{3.2}$$

where $\mathcal{L}_{\text{face}}$ and $\mathcal{L}_{\text{jaw}}$ are L2 distances between projected FLAME landmarks and detected landmarks, $\mathcal{L}_{\text{ear}}$ penalises ear landmark misalignment when ear landmarks are visible, and $\mathcal{L}_{\text{temp}}$ enforces temporal smoothness of camera rotation and translation across consecutive keyframes.

We assign per-region landmark weights to emphasise geometrically informative regions: eye contours receive weight 3.0, mouth contours 2.0, inner face landmarks 1.5, and jawline landmarks 0.5–0.75. The lower jawline weight reflects its sensitivity to occlusion and expression variation.

3.4.2 Stage 2: Joint Photometric and Landmark Fitting

The second stage jointly refines expression $\boldsymbol{\psi}$, jaw pose $\boldsymbol{\theta}_{\text{jaw}}$, eye pose, camera parameters, and optionally shape and texture. We optimise for 400 iterations

using Adam. The total loss combines photometric, landmark, and regularisation terms:

$$\mathcal{L}_{\text{key}} = \lambda_{\text{pho}}\mathcal{L}_{\text{pho}} + \lambda_{\text{lmk}}\mathcal{L}_{\text{lmk}} + \lambda_{\text{eyes}}\mathcal{L}_{\text{eyes}} + \lambda_{\text{ear}}\mathcal{L}_{\text{ear}} + \mathcal{L}_{\text{reg}}. \tag{3.3}$$

The photometric loss $\mathcal{L}_{\text{pho}}$ computes the pixel-wise L1 error between the rendered FLAME mesh and the input image within a face-masked region, weighted by $\lambda_{\text{pho}} = 2.0$. The landmark loss $\mathcal{L}_{\text{lmk}}$ computes a weighted L2 distance to the 68 detected landmarks, using the same per-region weights as Stage 1. The eye landmark loss $\mathcal{L}_{\text{eyes}}$ separately penalises eye landmark error to improve gaze accuracy.

The regularisation loss is:

$$\mathcal{L}_{\text{reg}} = \lambda_{\beta}\|\boldsymbol{\beta}\|^2 + \lambda_{\psi}\|\boldsymbol{\psi}\|^2 + \lambda_{\text{temp}}\|\boldsymbol{\psi}_k - \boldsymbol{\psi}_{k-1}\|^2, \tag{3.4}$$

where the first two terms penalise large shape and expression deviations from the mean, and the third term encourages temporal smoothness of expression parameters across consecutive keyframes.

Failure handling. When MediaPipe fails to detect landmarks on a keyframe, we initialise FLAME parameters from DECA and interpolate the camera pose and expression from neighbouring successful keyframes. This ensures continuous output even when individual frames are corrupted by motion blur or occlusion.

3.5 Pass 2: Intermediate Frame Fitting

Pass 2 fits FLAME parameters at intermediate timesteps between keyframe pairs using the B-spline landmark trajectories from preprocessing. This pass operates at higher temporal resolution than Pass 1 and produces the dense output sequence.

3.5.1 Camera Pose Interpolation

For each intermediate timestep $t \in (k, k+1)$, we linearly interpolate the camera rotation, translation, and field of view from the flanking keyframes k and $k+1$. The interpolated camera pose is held fixed during optimisation, ensuring geometric consistency with the keyframe fits.

3.5.2 Optimisation

At each intermediate timestep, we optimise expression $\boldsymbol{\psi}$, eye pose, and a learnable landmark offset $\Delta\mathbf{l} \in \mathbb{R}^2$ that compensates for systematic coordinate-space misalignment between the B-spline landmarks and the projected FLAME landmarks. We run 100 iterations of Adam with a learning rate of 0.01. The landmarks for each intermediate frame are queried from the B-splines at timestep t.

The total loss is:

$$\mathcal{L}_{\text{inter}} = \mathcal{L}_{\text{facial}} + \mathcal{L}_{\text{eyes}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{cm}}, \tag{3.5}$$

where:

- $\mathcal{L}_{\text{facial}} = 500 \cdot \|\hat{\mathbf{l}}_{17:68} - \mathbf{l}_{17:68}\|_1$ penalises L1 error on facial landmarks (indices 17–68, excluding the jawline),
- $\mathcal{L}_{\text{eyes}} = 500 \cdot \|\hat{\mathbf{l}}_{\text{eyes}} - \mathbf{l}_{\text{eyes}}\|_1$ penalises eye landmark error,
- $\mathcal{L}_{\text{reg}} = 0.025 \cdot \|\boldsymbol{\psi}\|^2 + 10.0 \cdot \|\Delta \mathbf{l}\|^2 + \lambda_{\text{temp}} \|\boldsymbol{\psi}_t - \boldsymbol{\psi}_{t-1}\|^2$ regularises expression parameters, penalises large landmark offsets, and encourages temporal smoothness,
- $\mathcal{L}_{\text{cm}}$ is an optional contrast maximisation loss (Section 3.6).

The exclusion of jawline landmarks (indices 0–16) from the facial loss reflects their lower reliability in ETAP tracking, as jaw contours lack the texture contrast needed for accurate event-based point tracking.

3.6 Contrast Maximisation

Contrast maximisation provides self-supervised refinement using event data, requiring no ground-truth labels. The core insight is that correct motion parameters warp events to a common reference time, producing a sharp Image of Warped Events (IWE), while incorrect parameters produce a blurred IWE [7]. This section describes the IWE construction, the two flow computation approaches we developed (2D and 3D), the event coordinate transform required for face-aligned crops, and the loss formulation. Figure 3.3 illustrates the overall pipeline.

3.6.1 Image of Warped Events

Given N events $\{(x_i, y_i, t_i, p_i)\}_{i=1}^N$ in a time window and a per-event warp function that maps each event to a reference time t_{ref}, we obtain warped positions (x'_i, y'_i) and accumulate them into an IWE via differentiable bilinear splatting:

$$\text{IWE}(x, y) = \sum_{i=1}^{N} p_i \cdot B(x - x'_i, y - y'_i), \tag{3.6}$$

where B is the bilinear interpolation kernel that distributes each event's polarity to the four nearest integer pixel locations with area-based weights. Figure 3.4 shows the effect on real data: an incorrect warp spreads events into a blurred IWE (low variance), while the correct warp aligns events into sharp edges (high variance).

3.6.2 Flow Computation

The warp function requires computing, for each event, where it would appear at t_{ref} given the FLAME-derived motion. We developed two approaches: a 2D approximation and an exact 3D method.

2D flow (initial approach). The 2D approach rasterises a dense per-pixel flow field by projecting FLAME vertices into normalised device coordinates (NDC) at timesteps t_0 and t_1 and computing the displacement:

$$\mathbf{u}(x, y) = \pi(\mathbf{v}_{t_1}) - \pi(\mathbf{v}_{t_0}), \tag{3.7}$$

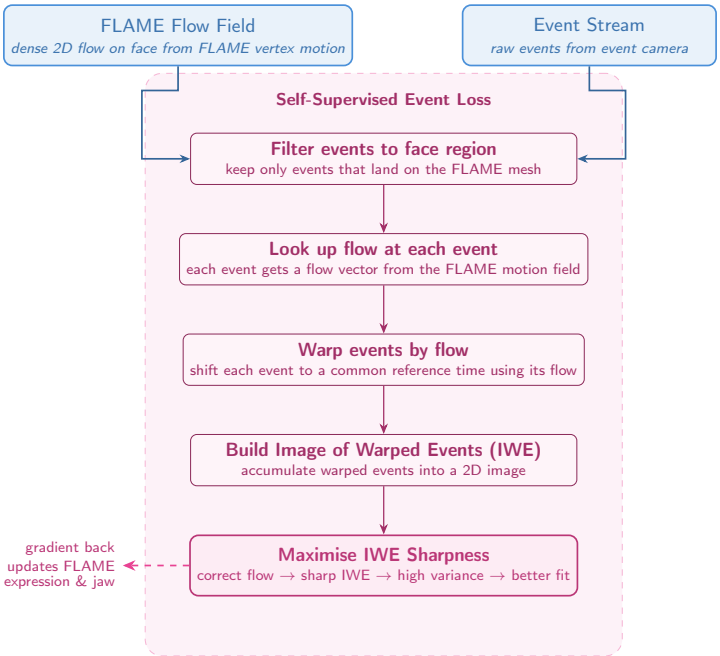


Figure 3.3: Contrast maximisation pipeline. The FLAME flow field and raw event stream are combined to produce an Image of Warped Events (IWE). Events are filtered to the face region, warped according to the FLAME-derived optical flow, and accumulated into the IWE. The loss maximises IWE variance, and gradients back-propagate to update FLAME expression and jaw parameters.

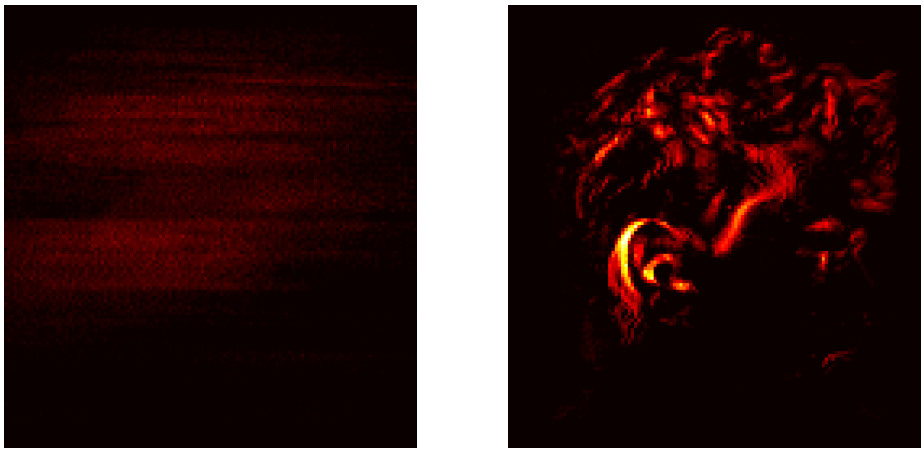
where π denotes camera projection and $\mathbf{v}_t$ is the vertex position at time t. Per-vertex displacements are rasterised into a dense flow map via barycentric interpolation, and each event looks up its flow from the map. Events are then warped linearly:

$$x'_i = x_i + u_x(x_i, y_i) \cdot (t_{\text{ref}} - t_i), \quad y'_i = y_i + u_y(x_i, y_i) \cdot (t_{\text{ref}} - t_i). \tag{3.8}$$

This linear approximation is fast but ignores perspective effects. The error is largest at the edges of the face mesh where depth discontinuities cause the 2D displacement to diverge from the true screen-space motion. In practice, the 2D approach produces IWE variance ratios near 1.0× (no improvement over zero flow), making it ineffective for optimisation.

3D flow (final approach). The 3D approach computes exact per-event warped positions by operating in camera space. For each event at pixel (x_i, y_i), we rasterise the FLAME mesh to determine which triangle face the event falls on, and recover the corresponding 3D camera-space surface point at timesteps t_0 and t_1, yielding $\mathbf{P}_0$ and $\mathbf{P}_1$. The warped 3D position at the reference time is:

$$\mathbf{P}_{\text{ref}} = \mathbf{P}_0 + \frac{t_{\text{ref}} - t_0}{t_1 - t_0}(\mathbf{P}_1 - \mathbf{P}_0), \tag{3.9}$$



(a) Incorrect motion (variance = 43.3) (b) Correct motion (variance = 253.3)

Figure 3.4: Image of Warped Events (IWE) on real event data from Subject Me. An incorrect warp velocity spreads events into horizontal streaks a, while the optimal velocity aligns events into sharp edges revealing facial structure b.

which is then projected back to screen coordinates via the camera intrinsics to obtain the warped position (x'_i, y'_i). Because the interpolation occurs in 3D before projection, this approach correctly handles perspective foreshortening and depth variation across the face. Gradients flow through the full chain—from IWE variance through the projection, 3D interpolation, and FLAME vertices back to the expression parameters. The 3D approach achieves 1.3–1.9× IWE variance ratios across subjects (Chapter 4), confirming that the FLAME motion model correctly explains the observed events. All experiments use the 3D flow approach.

3.6.3 Event Coordinate Transform

EFT operates on face-aligned image crops: each frame is transformed to a canonical 256×256 coordinate system via a face alignment pipeline. Events, which arrive in the original sensor coordinate system, must be transformed into this aligned space before CM can be applied.

We compute a 3×3 perspective homography $\mathbf{H}$ from the same alignment pipeline used for the image crop (shrink, crop, pad, perspective quad transform, output scale). Each event is transformed as:

$$\begin{pmatrix} x' \\ y' \\ w \end{pmatrix} = \mathbf{H} \begin{pmatrix} x_{\text{event}} \\ y_{\text{event}} \\ 1 \end{pmatrix}, \quad (x_{\text{aligned}}, y_{\text{aligned}}) = \left(\frac{x'}{w}, \frac{y'}{w} \right). \tag{3.10}$$

Events falling outside $[0, 256)^2$ after transformation are discarded. This ensures that events and the rendered FLAME mesh share the same coordinate system.

3.6.4 Loss Formulation

The contrast maximisation loss maximises the variance of the IWE:

$$\mathcal{L}_{\text{cm}} = -w_{\text{cm}} \cdot \text{Var}(\text{IWE}). \tag{3.11}$$

A high-variance IWE indicates that events are consistently warped to the same locations, which occurs when the estimated motion is correct. Events are filtered to the face region using the rendered FLAME mesh as a binary mask, ensuring that only face-related events contribute to the loss.

When used as an auxiliary loss alongside landmark supervision (CM+EFT), the CM term is added to the Pass 2 loss (Equation 3.5) with weight w_{cm}. When used as a standalone loss (CM-only), the landmark terms are removed and CM becomes the sole data-driven signal, with only expression regularisation and temporal smoothness as additional terms. Chapter 4 evaluates both configurations.

3.7 Implementation Details

We implement EFT in PyTorch [16] with differentiable rendering provided by PyTorch3D [18]. All fitting stages use the Adam optimiser [10]. The FLAME model consists of 5023 vertices with 300 identity shape parameters and 100 expression parameters. All FLAME fitting and rendering is performed at 256×256 resolution. Events are represented as mixed-density voxel stacks with 10 temporal bins and 20 000 events per stack.

Chapter 4

Experiments

This chapter evaluates EFT against four baselines on eight sequences spanning controlled and rapid facial motions. The core findings are: (1) EFT is the only event-based method that produces output on 100% of frames, while TimeLens fails on 50–63%; (2) EFT captures rapid motion—blinks and head rotation—that linear interpolation misses; (3) EFT produces the smoothest expression trajectories of all methods; and (4) contrast maximisation can fit a 3D FLAME model from events alone, without landmark supervision or regularisation.

We describe the evaluation protocol in Section 4.1, present overall landmark accuracy in Section 4.2, analyse TimeLens frame coverage and temporal coherence in Section 4.3, evaluate eye dynamics on rapid blinking in Section 4.4, examine rapid head motion in Section 4.5, report temporal properties in Section 4.6, and present a contrast maximisation ablation in Section 4.7.

4.1 Evaluation Protocol

4.1.1 Data

We evaluate on eight sequences with events simulated using vid2e [8] from four participants under two conditions: *standard* (S1–S4) with controlled, expression-specific motion, and *rapid* (R1–R4) with exaggerated facial dynamics. Table 4.1 summarises the sequence characteristics.

4.1.2 Held-out Frame Protocol

We evaluate temporal interpolation quality using a held-out frame protocol, illustrated in Figure 4.1. Given a sequence of N RGB frames, we retain every s-th frame as a keyframe and hold out the remaining $s-1$ intermediate frames. Each method receives only the keyframes and the full event stream, and must reconstruct FLAME parameters for each held-out frame. We evaluate at keyframe stride $s = 8$: the 7 intermediate frames between each pair of keyframes are held out and reconstructed.

Table 4.1: Evaluation sequences. Standard sequences (S1–S4) contain controlled single-expression motion; rapid sequences (R1–R4) contain fast, exaggerated motion.

ID	Condition	Primary motion
S1	Standard	Speech
S2	Standard	Smiling
S3	Standard	Blinking
S4	Standard	Jaw opening
R1	Rapid	Rapid blinking
R2	Rapid	Side-to-side head rotation
R3	Rapid	Diagonal jaw opening
R4	Rapid	Vertical jaw opening

4.1.3 Metrics

We evaluate using four complementary metrics:

FAN landmark error. Our primary metric uses the Face Alignment Network (FAN) [3] as an independent landmark detector. FAN is run on each original held-out frame to produce ground-truth 2D landmarks (indices 17–67, excluding the jawline). For each method, the reconstructed FLAME mesh is projected into the image plane and the mean Euclidean distance to the FAN landmarks is computed in pixels on a 256×256 image. FAN provides ground truth that is independent of both MediaPipe (used by EFT for keyframe detection) and DECA (used for per-frame fitting), avoiding circularity in the evaluation.

Eye Aspect Ratio (EAR). Aggregate landmark error cannot capture whether a method faithfully reproduces rapid, localised motions such as blinks. We therefore use the Eye Aspect Ratio (EAR) [19], a scalar metric widely adopted for blink detection in driver monitoring and fatigue analysis, which directly measures eye openness from six landmarks per eye:

$$\mathrm{EAR} = \frac{\|\mathbf{p}_2 - \mathbf{p}_6\| + \|\mathbf{p}_3 - \mathbf{p}_5\|}{2\,\|\mathbf{p}_1 - \mathbf{p}_4\|}, \tag{4.1}$$

where $\mathbf{p}_1$–$\mathbf{p}_6$ are six landmarks around each eye, extracted from MediaPipe's 478-point output. Open eyes produce EAR $\approx$ 0.3–0.4 and closed eyes EAR $\approx$ 0.1. Ground-truth EAR is computed from MediaPipe detections on the original frames; each method's EAR is computed from the corresponding FLAME mesh landmarks. We report the mean absolute error (MAE) between predicted and ground-truth EAR during automatically detected blink events (35th percentile threshold with hysteresis at the 50th percentile).

Expression velocity. We measure temporal smoothness as the mean L2 distance between consecutive expression parameter vectors. Lower values indicate smoother trajectories, which are desirable for animation applications.

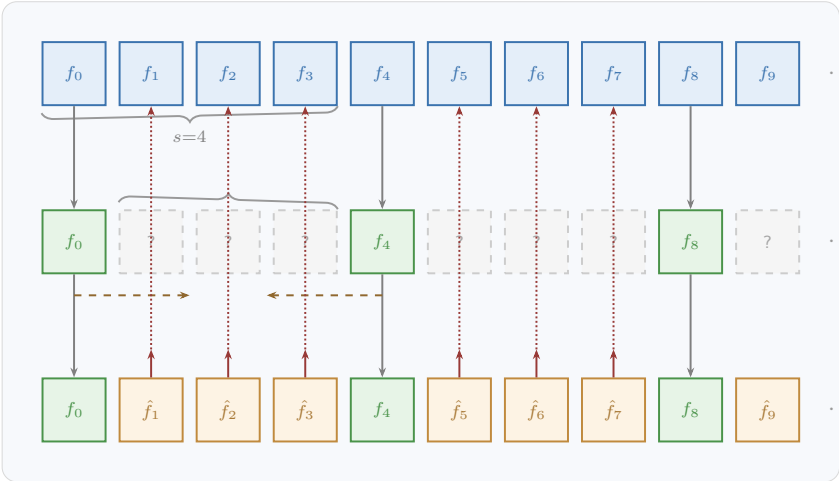


Figure 4.1: Held-out evaluation protocol. Top: all original frames with per-frame ground truth. Middle: only every s-th frame is retained as a keyframe; intermediate frames are held out. Bottom: each method reconstructs the held-out frames, which are compared against ground truth.

Expression dynamic range. We measure how much expression variation each method captures by computing the standard deviation of the expression parameter vector across all frames. Methods that faithfully track facial motion should produce dynamic range close to the ground truth.

4.1.4 Methods

We compare five methods:

- **EFT**: B-spline landmarks from bidirectional ETAP event tracking, fitted with the two-pass architecture (Chapter 3).
- **CM+EFT**: EFT with contrast maximisation as an auxiliary loss ($w_{\text{cm}} = 25$).
- **CM-only**: Contrast maximisation as the sole loss ($w_{\text{cm}} = 70$), with no landmark supervision.
- **Linear**: Linear interpolation of all FLAME parameters between the DECA fits of flanking keyframes.
- **TimeLens**: Event-based frame interpolation [20] followed by independent per-frame DECA fitting. Available for 7 of 8 sequences.

4.2 FAN Landmark Accuracy

Table 4.2 reports FAN landmark error across all eight subjects and five methods.

Table 4.2: FAN landmark error (px, 256 × 256) at stride 8. Bold indicates the best method per sequence. Linear interpolation achieves the lowest error on 7 of 8 sequences; EFT achieves the lowest on R2.

Sequence	EFT	CM+EFT	CM-only	Linear	TimeLens	Best
S1	5.5	4.0	3.7	**3.5**	5.5	Linear
S2	5.8	4.4	4.1	**3.9**	5.4	Linear
S3	5.6	4.0	3.8	**3.6**	—	Linear
S4	5.4	4.0	3.5	**3.3**	4.7	Linear
R1	5.4	5.0	4.4	**3.6**	4.6	Linear
R2	**21.1**	21.5	21.6	31.2	13.3	TimeLens
R3	6.5	5.6	5.3	**4.8**	5.1	Linear
R4	5.2	4.9	4.9	**4.4**	5.7	Linear

Linear interpolation achieves the lowest FAN error on 7 of 8 sequences. This reflects an inherent advantage: Linear interpolates DECA FLAME parameters between keyframes, and since DECA was fitted to the original RGB frames, the interpolated output remains close to the ground-truth fits.

The exception is R2 (rapid head rotation), where the participant performs fast side-to-side head rotation. Linear interpolation fails to approximate the nonlinear head trajectory and produces 31.2 px error—50% higher than EFT's 21.1 px. TimeLens achieves the lowest error on R2 (13.3 px), but this result is subject to survivorship bias, as discussed in Section 4.3.

4.3 TimeLens Analysis

TimeLens synthesises intermediate RGB frames from boundary keyframes and the event stream, then runs DECA independently on each synthesised frame. This section examines TimeLens's frame coverage, the resulting survivorship bias in its reported accuracy, and its temporal coherence.

4.3.1 Frame Coverage

TimeLens fails to reconstruct 50–63% of intermediate frames across sequences. DECA either fails to detect a face in the synthesised frame or produces a degenerate fit. Figure 4.2 shows a typical failure on S1: TimeLens produces severe visual artefacts in the interpolated frame, causing DECA to fit a degenerate mesh (NME = 0.53), while EFT reconstructs the correct expression (NME = 0.07). Table 4.3 reports coverage and FAN error under two conditions: using only the frames where TimeLens succeeds, and using linear interpolation as fallback for failed frames.

TimeLens's headline FAN errors in Table 4.2 reflect only the frames where it succeeded—the easier subset. On R2, TimeLens's 13.3 px error comes from only 37% of frames (32 of 87). With linear interpolation as fallback for the remaining 63%, TimeLens achieves 18.4 px—still better than EFT's 21.1 px, but far from the reported 13.3 px.

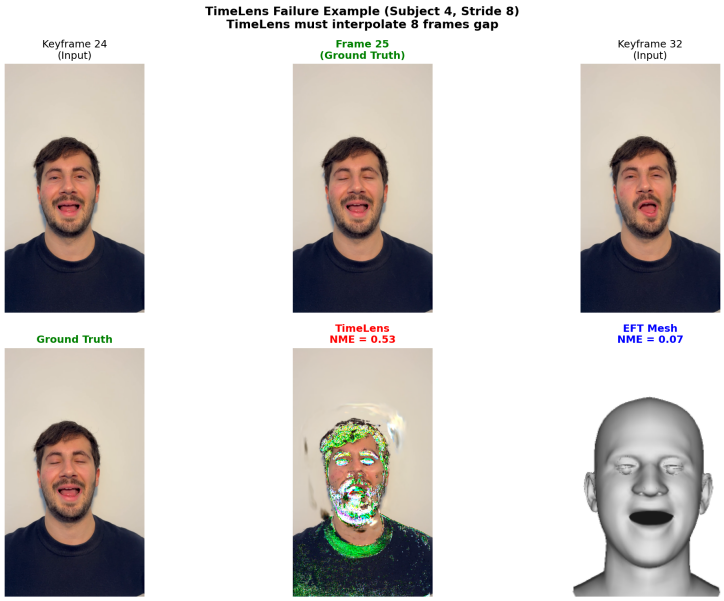


Figure 4.2: TimeLens failure example on S1 at stride 8. Top row: boundary keyframes and ground-truth intermediate frame. Bottom row: ground truth, TimeLens output with severe artefacts (NME = 0.53), and EFT mesh reconstruction (NME = 0.07). TimeLens's synthesised frame contains corruption that causes DECA to produce a degenerate fit.

Table 4.4 breaks down R2 by frame subset.

On the 63% of frames where TimeLens fails, EFT achieves 14.4 px. EFT is the only event-based method that produces output on 100% of frames without requiring a fallback.

4.3.2 Temporal Coherence

Because TimeLens fits each synthesised frame independently, it produces temporally incoherent expression trajectories. As reported in Section 4.6.1, TimeLens exhibits 2–8× higher expression velocity than EFT on every sequence. This jitter makes TimeLens unsuitable for applications requiring temporally coherent output, such as facial animation. EFT's B-spline interpolation enforces temporal continuity by construction, producing smooth expression changes without the discontinuities inherent in independent per-frame fitting.

4.4 Rapid Blink: Eye Dynamics

This section evaluates how faithfully each method captures rapid eye dynamics using the EAR metric on sequence R1 (rapid blinking), which contains 15 automatically detected blink events over 106 frames. Linear interpolation smooths through rapid blink transitions at stride 8, while EFT's event-based tracking captures the timing and depth of each blink.

Table 4.3: TimeLens frame coverage at stride 8. TimeLens fails on 50–63% of intermediate frames. FAN error is reported on successful frames only and with linear interpolation fallback for failed frames.

		TimeLens			
Sequence	Coverage	Success	Fallback	EFT	Linear
S1	48%	5.5	4.4	5.5	**3.5**
S2	50%	5.4	—	5.8	**3.9**
S4	49%	4.7	—	5.4	**3.3**
R1	49%	4.6	—	5.4	**3.6**
R2	37%	**13.3**	18.4	21.1	31.2
R3	46%	5.1	—	6.5	**4.8**
R4	51%	5.7	—	5.2	**4.4**

Table 4.4: R2 (rapid head rotation) FAN error (px) by frame subset. TimeLens excels on its successful frames (37%) but produces no output on the remaining 63%. EFT works on all frames.

Frame subset	EFT	TimeLens	Linear
TL-success (32/87)	33.2	**13.3**	48.0
TL-failed (55/87)	**14.4**	—	21.4
All frames (87/87)	**21.1**	18.4 (fallback)	31.2

4.4.1 Per-Event Analysis

Table 4.5 reports the EAR MAE for each detected blink event.

EFT achieves the lowest EAR error on 11 of 15 blink events (73%), with a mean MAE 16% lower than linear interpolation (0.089 vs. 0.106). The largest advantages occur on events 5 and 6 (+34–38%), where the blinks are rapid and linear interpolation smooths through them entirely.

4.4.2 Cross-Subject EAR

Table 4.6 reports EAR MAE during blink events across all eight sequences.

EFT wins on 5 of 8 sequences, including all sequences with significant eye dynamics (S1, S2, S3, R1, R2). Linear interpolation wins on sequences where blink events are incidental to the primary motion (S4, R3, R4), where EAR variations are small and linear parameter interpolation approximates them well.

4.5 Rapid Head Motion

Sequence R2 involves fast side-to-side head rotation, producing nonlinear head trajectories that linear interpolation cannot approximate. Both EFT and CM-only substantially outperform linear interpolation on this sequence, with EFT achieving the lowest error among methods that work on all frames.

Table 4.5: EAR MAE per blink event on R1 at stride 8. Bold indicates the lowest error per event. Δ: EFT improvement over Linear (positive = EFT better). EFT achieves the lowest error on 11 of 15 events.

Event	Frames	EFT	CM+EFT	Linear	Δ
1	6–10	**0.054**	0.097	0.072	+25%
2	11–15	0.085	**0.075**	0.066	−29%
3	18–21	0.149	**0.140**	0.171	+13%
4	25–28	**0.095**	0.128	0.118	+20%
5	32–35	**0.056**	0.110	0.090	+38%
6	38–41	**0.068**	0.103	0.103	+34%
7	45–48	0.143	**0.124**	0.155	+7%
8	52–55	0.119	**0.104**	0.132	+9%
9	57–60	**0.086**	0.117	0.101	+16%
10	64–67	**0.085**	0.138	0.117	+27%
11	70–74	**0.067**	0.083	0.085	+21%
12	78–82	**0.048**	0.098	0.066	+27%
13	85–88	**0.090**	0.110	0.094	+4%
14	92–96	**0.125**	0.134	0.136	+8%
15	101–106	**0.071**	0.095	0.085	+17%
Mean MAE		**0.089**	0.110	0.106	+16%

4.5.1 Per-Interval Analysis

Table 4.7 breaks down FAN error by keyframe interval.

EFT wins 10 of 13 intervals, with the largest advantages during peak head rotation (intervals 8–32), where the nonlinear trajectory causes linear interpolation error to exceed 60 px. CM-only closely matches EFT throughout, confirming that contrast maximisation captures the side-to-side motion through its 3D flow field (Section 3.6). Linear interpolation wins only on the final three intervals (80–99), where the head is nearly stationary.

4.5.2 Qualitative Results

Figure 4.3 shows three intervals during peak head rotation. EFT tracks the head pose accurately, while linear interpolation under-rotates the mesh.

4.6 Temporal Properties

This section evaluates the temporal quality of reconstructed expression sequences. EFT produces the smoothest trajectories on the standard sequences and captures expression dynamic range closest to the ground truth on the rapid sequences.

4.6.1 Expression Velocity

Table 4.8 reports expression velocity across all subjects and methods.

Table 4.6: EAR MAE during blink events across all sequences. EFT captures blink dynamics more accurately on 5 of 8 sequences. Linear wins on sequences where eye motion is incidental to the primary motion.

Sequence	EFT	CM+EFT	CM-only	Linear	Best
S1	**0.059**	0.069	0.070	0.068	EFT
S2	**0.033**	0.058	0.056	0.062	EFT
S3	**0.045**	0.051	0.056	0.057	EFT
S4	0.049	0.046	0.041	**0.028**	Linear
R1	**0.113**	0.124	0.127	0.127	EFT
R2	**0.021**	0.069	0.079	0.065	EFT
R3	0.075	0.059	0.050	**0.048**	Linear
R4	0.090	0.027	0.024	**0.009**	Linear

EFT produces the smoothest expression trajectories on all four standard sequences, with B-spline interpolation of ETAP trajectories enforcing temporal continuity by construction. On the four rapid sequences, linear interpolation is smoothest because it interpolates directly between keyframe parameters—but this smoothness comes at the cost of under-representing expression variation (Section 4.6.2).

CM+EFT and CM-only produce substantially higher expression velocity (5–8× EFT), reflecting the high-frequency expression changes driven by the contrast maximisation loss. TimeLens is the least smooth method on every sequence where it is available, with 2–8× higher expression velocity than EFT.

4.6.2 Expression Dynamic Range

Table 4.9 reports the standard deviation of each method's expression vector compared to the ground truth.

On all four rapid sequences, EFT captures expression dynamic range closest to the ground truth. Linear interpolation under-represents expression variation by 30–50% on the rapid sequences (e.g. 14.4 vs. GT 18.8 on R1, 22.4 vs. GT 31.4 on R2), confirming that it smooths through rapid motions. CM+EFT and CM-only over-represent dynamic range by 50–200%, reflecting unconstrained expression changes driven by the contrast maximisation loss.

On the four standard sequences, where motions are slower, linear interpolation is closest to the ground truth. This is consistent with slow, controlled motions being well-approximated by linear parameter interpolation.

4.7 Contrast Maximisation

This section evaluates contrast maximisation (CM) as a loss component. We validate the signal quality through IWE variance ratios and demonstrate that CM can fit FLAME parameters without landmark supervision or expression regularisation.

Table 4.7: R2 (rapid head rotation) FAN error (px) by keyframe interval. Δ: EFT improvement over Linear. EFT wins 10 of 13 intervals, with the largest advantages during peak rotation (intervals 8–32).

Interval	n	EFT	CM-only	Linear	Δ
0–8	7	**20.1**	20.3	26.8	+6.7
8–16	7	**47.2**	47.2	74.4	+27.3
16–24	7	51.0	**50.2**	83.4	+32.4
24–32	7	37.0	**36.9**	64.1	+27.2
32–40	7	**23.7**	24.2	38.2	+14.5
40–48	7	**24.7**	26.5	28.3	+3.6
48–56	7	**10.1**	10.6	15.6	+5.5
56–64	7	**16.4**	17.7	21.6	+5.2
64–72	7	**10.7**	11.8	12.2	+1.4
72–80	7	**11.8**	12.6	12.3	+0.6
80–88	7	5.9	6.6	**5.3**	−0.7
88–96	7	4.7	5.2	**3.8**	−0.9
96–99	3	4.4	4.2	**3.4**	−1.0
Mean		**20.6**	21.1	30.0	+9.4

4.7.1 IWE Variance

Table 4.10 reports the IWE variance ratio: the factor by which the FLAME-derived optical flow improves IWE sharpness over zero flow.

The FLAME-derived flow field improves IWE variance by 1.3–1.8× across all sequences, confirming that the FLAME motion model correctly explains the observed events when the flow is computed with the 3D warp mode (Section 3.6).

4.7.2 CM Without Regularisation

To test whether contrast maximisation provides sufficient constraint without any prior, we run CM-only with zero expression regularisation. Table 4.11 compares the results.

Removing expression regularisation entirely produces less than 0.2 px difference in FAN error. This demonstrates that contrast maximisation alone provides sufficient constraint to fit a 3D FLAME model from events, without any landmark supervision or regularisation. The physics of event generation—brightness changes at edges caused by motion—implicitly constrains the optimisation to produce plausible facial expressions. The IWE variance loss surface is well-behaved enough that the optimiser does not diverge even with full freedom over the 100-dimensional expression vector.

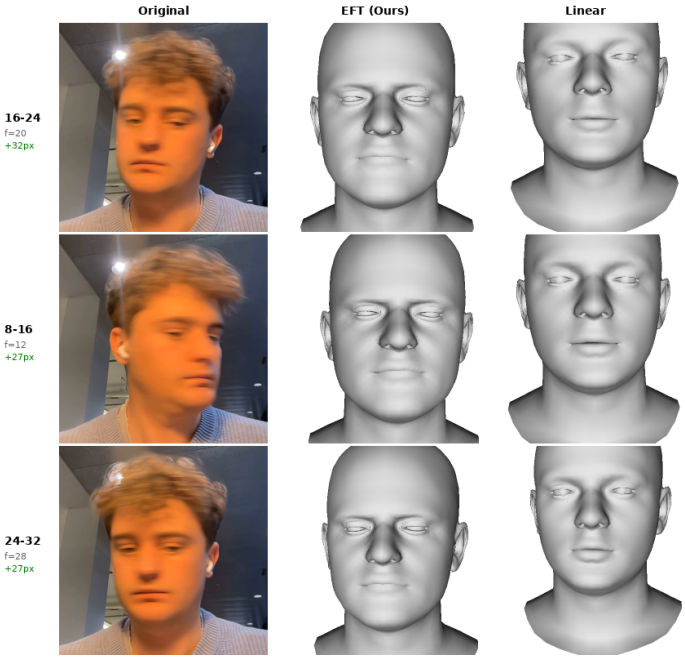


Figure 4.3: Qualitative comparison on R2 during peak rotation intervals. Left: original frame. Centre: EFT. Right: linear interpolation. EFT tracks the nonlinear head trajectory while linear interpolation under-rotates the mesh.

Table 4.8: Expression velocity (mean L2 between consecutive expression vectors; lower = smoother). Bold indicates the smoothest method per sequence. EFT is smoothest on all four standard sequences and smoother than TimeLens on every sequence.

Sequence	EFT	CM+EFT	CM-only	Linear	TimeLens
S1	**0.65**	5.38	5.70	0.69	1.92
S2	**0.75**	5.05	5.09	0.82	2.19
S3	**0.86**	6.09	6.15	0.86	—
S4	**0.69**	5.45	5.33	0.81	1.75
R1	1.38	6.10	5.97	**0.42**	2.87
R2	1.97	7.14	7.26	**0.64**	5.26
R3	1.18	4.79	5.18	**0.41**	2.26
R4	2.06	7.68	7.72	**0.52**	2.70

Table 4.9: Expression dynamic range (std of expression vector). The method closest to the ground-truth (GT) value best represents the true range of facial motion. EFT is closest on all four rapid sequences; Linear is closest on the standard sequences.

Sequence	EFT	CM+EFT	CM-only	Linear	GT	Closest
S1	22.6	57.0	58.9	37.1	43.8	Linear
S2	26.8	62.8	63.7	45.1	52.4	Linear
S3	28.0	68.2	68.6	46.6	52.3	Linear
S4	22.1	57.6	56.5	31.5	38.6	Linear
R1	**18.3**	51.4	51.8	14.4	18.8	EFT
R2	**35.4**	64.5	65.9	22.4	31.4	EFT
R3	**21.4**	46.2	49.1	11.3	16.7	EFT
R4	**26.6**	62.2	63.0	13.5	20.5	EFT

Table 4.10: IWE variance ratio (higher = better event alignment). The FLAME flow field consistently improves IWE sharpness, confirming that expression changes produce optical flow consistent with the observed events.

Sequence	Events/interval	CM+EFT	CM-only
S1	151k	1.77×	1.74×
S2	130k	1.43×	1.32×
S3	81k	1.57×	1.53×
S4	39k	1.55×	1.49×

Table 4.11: CM-only with and without expression regularisation. FAN error is nearly identical (<0.2 px difference), demonstrating that the CM loss alone constrains the 100-dimensional expression vector without requiring priors.

Sequence	With reg	No reg	Δ
S1	3.96	3.96	0.00
S2	4.32	4.32	0.00
S3	4.32	4.33	+0.01
S4	4.10	4.10	0.00
R1	5.14	5.11	−0.03
R3	5.67	5.47	−0.20

Chapter 5

Discussion

Explain both the advantages and limitations of your approach.

5.1 Conclusion

Summarize your work and what came out of it.

5.2 Future Work

How would you extend the work? Can you propose another approach?

Appendix A

Something

In the appendix, you can provide some more data, a tutorial on how to run your code, a detailed proof, etc.

It is, however, not a requirement to have an appendix.

Bibliography

[1] Thabo Beeler, Fabian Hahn, Derek Bradley, Bernd Bickel, Paul Beardsley, Craig Gotsman, Robert W. Sumner, and Markus Gross. High-quality passive facial performance capture using anchor frames. *ACM Transactions on Graphics (Proceedings of SIGGRAPH)*, 30(4):75:1–75:10, 2011. doi: 10.1145/2010324.1964970.

[2] Volker Blanz and Thomas Vetter. A morphable model for the synthesis of 3D faces. In *Proceedings of the 26th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pages 187–194. ACM Press/Addison-Wesley, 1999.

[3] Adrian Bulat and Georgios Tzimiropoulos. How far are we from solving the 2D & 3D face alignment problem? (and a dataset of 230,000 3D facial landmarks). In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1021–1030, 2017.

[4] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4690–4699, 2019.

[5] Carl Doersch, Ankush Gupta, Larisa Markeeva, Adria Recasens, Lucas Smaira, Yusuf Aytar, Joao Carreira, Andrew Zisserman, and Yi Yang. TAP-Vid: A benchmark for tracking any point in a video. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 13610–13626, 2022.

[6] Yao Feng, Haiwen Feng, Michael J. Black, and Timo Bolkart. Learning an animatable detailed 3D face model from in-the-wild images. *ACM Transactions on Graphics (ToG), Proc. SIGGRAPH*, 40(4):88:1–88:13, 2021.

[7] Guillermo Gallego, Henri Rebecq, and Davide Scaramuzza. A unifying contrast maximization framework for event cameras, with applications to motion, depth, and optical flow estimation. In *IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, pages 3867–3876, 2018. doi: 10.1109/CVPR.2018.00407.

[8] Daniel Gehrig, Mathias Gehrig, Javier Hidalgo-Carrió, and Davide Scaramuzza. Video to Events: Recycling video datasets for event cameras. In *IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2020. doi: 10.1109/cvpr42600.2020.00364.

[9] Friedhelm Hamann, Daniel Gehrig, Filbert Febryanto, Kostas Daniilidis, and Guillermo Gallego. Etap: Event-based tracking of any point. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 27186–27196, 2025.

[10] Diederik P. Kingma and Jimmy L. Ba. Adam: A method for stochastic optimization. *Int. Conf. Learn. Representations (ICLR)*, 2015.

[11] Takuya Kohyama, Tharindu Munasinghe, and Shintaro Sonoda. 3D human scan with a moving event camera. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 2661–2670, 2024.

[12] Tianye Li, Timo Bolkart, Michael J. Black, Hao Li, and Javier Romero. Learning a model of facial shape and expression from 4D scans. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia)*, 36(6):194:1–194:17, 2017. URL `https://doi.org/10.1145/3130800.3130813`.

[13] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: A skinned multi-person linear model. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia)*, 34(6):248:1–248:16, 2015.

[14] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. Mediapipe: A framework for perceiving and processing reality. In *Third Workshop on Computer Vision for AR/VR at IEEE Computer Vision and Pattern Recognition (CVPR) 2019*, 2019. URL `https://mixedreality.cs.cornell.edu/s/NewTitle_May1_MediaPipe_CVPR_CV4ARVR_Workshop_2019.pdf`.

[15] Yongrui Ma, Shi Guo, Yutian Chen, Tianfan Xue, and Jinwei Gu. TimeLens-XL: Real-time event-based video frame interpolation with large motion. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024.

[16] Adam Paszke, Sam Gross, Soumith Chintala, Gregory Chanan, Edward Yang, Zachary DeVito, Zeming Lin, Alban Desmaison, Luca Antiga, and Adam Lerer. Automatic differentiation in PyTorch. In *NIPS Workshops*, 2017.

[17] Pascal Paysan, Reinhard Knothe, Brian Amberg, Sami Romdhani, and Thomas Vetter. A 3D face model for pose and illumination invariant face recognition. In *Proceedings of the 6th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, pages 296–301. IEEE, 2009.

[18] Nikhila Ravi, Jeremy Reizenstein, David Novotny, Taylor Gordon, Wan-Yen Lo, Justin Johnson, and Georgia Gkioxari. Accelerating 3D deep learning with PyTorch3D. In *SIGGRAPH Asia 2020 Courses*. ACM, 2020.

[19] Tereza Soukupová and Jan Čech. Real-time eye blink detection using facial landmarks. In *21st Computer Vision Winter Workshop (CVWW)*, pages 1–8, 2016.

[20] Stepan Tulyakov, Daniel Gehrig, Stamatios Georgoulis, Julius Erbach, Mathias Gehrig, Yuanyou Li, and Davide Scaramuzza. TimeLens: Event-based video frame interpolation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4417–4426, 2021.

[21] Stepan Tulyakov, Alfredo Bochicchio, Daniel Gehrig, Stamatios Georgoulis, Yuanyou Li, and Davide Scaramuzza. Time Lens++: Event-based frame interpolation with parametric non-linear flow and multi-scale fusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17451–17460, 2022.

[22] Lan Xu, Weipeng Xu, Vladislav Golyanik, Marc Habermann, Lu Fang, and Christian Theobalt. EventCap: Monocular 3D capture of high-speed human motions using an event camera. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4968–4978, 2020.

[23] Yingyan Xu, Prashanth Chandran, Sebastian Weiss, Markus Gross, Gaspard Zoss, and Derek Bradley. Artist-friendly relightable and animatable neural heads. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2457–2467, 2024. doi: 10.1109/CVPR52733.2024.00238.

[24] Wojciech Zielonka, Timo Bolkart, and Justus Thies. Towards metrical reconstruction of human faces. In *European Conference on Computer Vision (ECCV)*, pages 250–269. Springer, 2022.

ROBOTICS &
PERCEPTION
GROUP

Title of work:

Event-based Facial Tracking

Thesis type and date:

Master Thesis, March 2026

Supervision:

Roberto Pellerito
Nikola Zubić
Prof. Dr. Davide Scaramuzza

Student:

Name:	Samuel Francis Wallace
E-mail:	samuelfrancis.wallace@uzh.ch
Legi-Nr.:	23-741-440

Statement regarding plagiarism:

By signing this statement, I affirm that I have read the information notice on plagiarism, independently produced this paper, and adhered to the general practice of source citation in this subject-area.

Information notice on plagiarism:

`https://www.ius.uzh.ch/dam/jcr:00000000-100f-6f7c-ffff-ffff869554ef/LK_Plagiarism.pdf`

Zurich, 21. 3. 2026: _____